

Phishing Detection and Awareness Report

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Future Interns Cyber Security Internship Task 2

Executive Summary

A phishing email analysis was conducted to identify common indicators used in email-based social engineering attacks to assess the associated risks to users and organizations. The assessments involved reviewing samples, examining sender information, analyzing message content, and identifying characteristics commonly associated with phishing campaigns.

The analysis identified multiple phishing indicators, including brand impersonation, suspicious sender information, deceptive links, unexpected attachments, and urgency-based messaging designed to pressure recipients into taking immediate action. Based on the findings, the analyzed emails were classified according to risk level and likelihood of malicious intent.

The results demonstrate how attackers exploit trusted brands and common business processes, such as document sharing, voicemail notifications, and fax delivery services, to deceive users into revealing sensitive information or interacting with malicious content. These attackers can lead to credential theft, malware infections, financial loss, and authorized access to organizational systems.

Overall, the assessment highlights the importance of user awareness as a critical defense against phishing attacks. By recognizing common phishing indicators and following recommended security practices, users can significantly reduce the risk of becoming victims of email-based cyber threats.

Scope and Methodology

Scope

This assessment focused on analyzing email samples to identify phishing indicators and evaluate the potential risks they pose to users and organizations. The review included examination of sender information, email content, branding, attachments, embedded links, and social engineering techniques commonly used in phishing campaigns.

Included Activities

- Analysis of phishing email samples
- Review of sender names and email addresses
- Inspection of email content and messaging
- Identification of phishing indicators
- Risk classification (Safe, Suspicious, or Phishing)
- Evaluation of potential business impact
- Development of user awareness and prevention guidelines

Excluded Activities

- Opening suspicious attachments
- Clicking potentially malicious links
- Active interaction with phishing websites
- Malware execution or testing
- Unauthorized access attempts
- Any activity that could compromise systems or data

Methodology

The assessment was conducted using a structured email analysis approach commonly used by Security Operations Center (SOC) analysts.

Step 1: Email Collection:

Publicly available phishing email samples were collected from educational and research sources for analysis.

Step 2: Sender Verification:

Sender names, email addresses, and domains were reviewed to identify signs of spoofing, impersonation, or suspicious domain usage.

Step 3: Content Analysis:

Email subject lines, message content, branding, and calls to action were examined for common phishing techniques such as urgency, fear-based language, and impersonation of trusted organizations.

Step 4: Link and Attachment Review:

Links and attachments referenced within the emails were assessed for indicators of malicious intent without interacting with or executing any content.

Step 5: Risk Classification

Each email was classified as Safe, Suspicious, or Phishing based on the indicators identified during analysis.

Step 6: Awareness Recommendations

Practical security awareness guidance and prevention measures were developed to help users recognize and avoid phishing attempts.

Tools Used

Tool	Purpose
Email Samples Repository	Source of phishing email examples
Google Message Header Analyzer	Email header analysis
MXToolbox Email Header Analyzer	Sender and header verification
Web Browser Tools	Link and domain inspection
Microsoft Word	Report creation and documentation

This methodology ensured that the assessment remained safe, ethical, and focused on awareness and risk identification rather than exploitation.

Sample Analysis

Email 1 Analysis: Fake Office 365 Voicemail Notification

Classification: Phishing

Risk Level: High

Threat Type: Credential Harvesting

Indicators Identified

Indicator	Observation
Brand Impersonation	Uses Microsoft Office 365 branding
Urgency	Claims a new voicemail required immediate action
Unexpected Communication	Most users are not expecting voicemail notification via email
Attachment Request	Encourages opening a file to listen to a message
Social Engineering	Uses a phone number and voicemail details to appear legitimate

Why Is It Suspicious?

The email attempts to convince the recipient that a voicemail has been received and that an attachment must be downloaded to listen to it. Attackers commonly use fake voicemail notification because users are curious and often click without verifying the sender.

Potential Impact

- Malware infection
- Credential theft
- Account compromise
- Unauthorized access to company resources

Email Analysis 2: Fake SharePoint Document Notification

Classification: Phishing

Risk Level: High

Threat Type: Credential Harvesting

Indicators Identified

Indicator	Observation
Brand Impersonation	Uses SharePoint branding
Fake Document Alert	Claims a document is waiting to be reviewed
Call-to-Action Button	"Preview Fax Document Here" button designed to encourage clicks
Unusual File Description	Includes unnecessary technical details to appear legitimate
Unexpected Document	Recipient did not request or expect the file

Why Is It Suspicious?

Attackers frequently impersonate SharePoint because employees trust Microsoft services. The primary objective is usually to redirect users to a fake login page where credentials can be stolen.

Potential Impact

- Microsoft 365 account compromise
- Password theft
- Unauthorized access to company files
- Business email compromise (BEC)

Email 3 Analysis: Future Interns Internship Submission Email

Classification: Safe

Risk Level: Low

Threat Type: None

Summary

The email appears to be a legitimate communication from Future Interns regarding internship submission procedures. The message contains clear organizational branding, a legitimate sender address, and information that is relevant to the recipient's internship activities.

Indicators Identified

Indicator	Observation
Verified Branding	Consistent Future Interns branding and contact information
Relevant Content	Relates directly to internship participation
Legitimate Sender	Sender address matches the organization name
Professional Communication	Clear structure and professional language
Expected Action	Internship submission request aligns with program activities

Why It Appears Legitimate?

The email references an expected internship process and provides information relevant to participants. Unlike phishing emails, it does not attempt to create excess fear, request passwords, ask for OTPs, or impersonate an unrelated organization.

Potential Risk

No significantly security concerns were identified during the review. Users should still verify links before clicking as a general security best practice.

Awareness Lesson Learned

These samples demonstrate the three most common phishing lures used in organizations:

1. Voicemail Notification - Pretend a message requires immediate preview.
2. Document Sharing Requests - Impersonate Microsoft SharePoint or OneDrive.
3. Fax/Invoice Deliveries - Use attachments to distribute malware.

User Guidance

Before interacting with any email:

- Verify the sender address
- Confirm you were expecting the message
- Hover over links before clicking
- Verify requests through official channels
- Report suspicious emails to IT/Security teams

What To Avoid:

- Do not open unexpected attachments
 - Do not enter credentials after clicking email links
 - Do not trust branding alone
 - Do not respond to suspicious messages
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Conclusion

The phishing email assessment identified multiple phishing indicators, including brand impersonation, suspicious attachments, and urgency-based messaging designed to manipulate recipients into taking immediate action. All analyzed samples were classified as phishing due to their high potential for credential theft, malware infection, and unauthorized access.

The findings highlight the importance of user awareness in preventing phishing attacks. By verifying sender information, avoiding suspicious links and attachments, and reporting suspicious emails, users can significantly reduce the risk of falling victim to email-based cyber threats.